

Montgomery College
Rockville Campus



FINAL PROJECT ANALYSIS FOR HONDA PILOT



May 18, 2025

Paula Buitrago, Christine Saint-Firmin, Ayomide Joe-Adigwe
Professor Angel Salinas

Introduction to Business Analytics: BSAN 101

Table of Contents

<i>Introduction:</i>	2
<i>Honda's Financial Statement:</i>	3
<i>Honda's Financial Statement Analysis:</i>	4
<i>Monthly Descriptive Analysis of Honda Pilot:</i>	4
<i>Annual Descriptive Analysis of Honda Pilot:</i>	6
<i>Forecasting Analysis:</i>	8
<i>References</i>	10

Introduction:

Over the past decade the American automotive industry has undergone significant transformations shaped by technological advances, consumer trends, and macroeconomic imbalances like the Covid-19 pandemic. As part of this reformation, Honda Motor Co Ltd has realigned its strategy by restructuring operations and investing in innovation while prioritizing financial stability. Determining the current financial status is crucial for analyzing its competitive standing and forecasting future outcomes.

The most recent financial statement from Honda will be reviewed in this research paper, which will entail of a summary and interpretation of its balance sheet and cash flow activities. The descriptive and forecasting analyses that were obtained from the monthly and annual sales data will thereafter be our main emphasis. Utilizing the statistical techniques taught in class, such as the exponential smoothing models and moving average, we will evaluate sales trends and generate short-term forecasts. The paper will conclude with a brief reflection on Honda's financial and sales performance, along with recommendations based on the forecasting analysis.

Honda's Financial Statement:

March 31, 2023 and 2024

		Yen (millions)	
		Mar. 31, 2023	Mar. 31, 2024
Assets			
Current assets:			
Cash and cash equivalents	3,803,014	4,954,565	
Trade receivables	1,060,271	1,240,090	
Receivables from financial services	1,899,493	2,558,594	
Other financial assets	263,892	229,583	
Inventories	2,167,184	2,442,969	
Other current assets	384,494	446,763	
Total current assets	9,578,348	11,872,564	
Non-current assets:			
Investments accounted for using the equity method	915,946	1,206,968	
Receivables from financial services	3,995,259	5,616,676	
Other financial assets	855,070	968,142	
Equipment on operating leases	4,726,292	5,202,768	
Property, plant and equipment	3,168,109	3,234,413	
Intangible assets	870,900	999,689	
Deferred tax assets	105,792	170,856	
Other non-current assets	454,351	502,074	
Total non-current assets	15,091,719	17,901,586	
Total assets	24,670,067	29,774,150	
Liabilities and Equity			
Current liabilities:			
Trade payables	1,426,333	1,609,836	
Financing liabilities	3,291,195	4,105,590	
Accrued expenses	419,570	638,319	
Other financial liabilities	324,110	340,858	
Income taxes payable	86,252	157,410	
Provisions	362,701	566,722	
Other current liabilities	741,963	904,757	
Total current liabilities	6,652,124	8,323,492	
Non-current liabilities:			
Financing liabilities	4,373,973	6,057,967	
Other financial liabilities	288,736	316,919	
Retirement benefit liabilities	255,852	284,844	
Provisions	270,169	385,001	
Deferred tax liabilities	877,300	855,067	
Other non-current liabilities	449,622	544,988	
Total non-current liabilities	6,515,652	8,444,786	
Total liabilities	13,167,776	16,768,278	
Equity:			
Common stock	86,067	86,067	
Capital surplus	185,589	205,073	
Treasury stock	(484,931)	(550,808)	
Retained earnings	9,980,128	10,644,213	
Other components of equity	1,417,397	2,312,450	
Equity attributable to owners of the parent	11,184,250	12,696,995	
Non-controlling interests	318,041	308,877	
Total equity	11,502,291	13,005,872	
Total liabilities and equity	24,670,067	29,774,150	

Honda's Financial Statement Analysis:

All of Honda's key financial metrics showed robust growth for the fiscal year that ended on March 31, 2024. Due in large part to higher car sales and advantageous foreign exchange rates, the company's consolidated sales revenue increased by 20.8% to reach JPY 20,428.8 billion. Despite rising expenses, operating profit increased significantly by 77.0% (to JPY 1,381.9 billion), primarily because of improved pricing strategies, cost control, and higher sales volumes.

The net profit attributable to owners increased by 70.0% to JPY 1,107.1 billion, while the profit before taxes increased by 86.7%. As a result of a three-for-one stock split that was recorded retroactively, Honda's earnings per share (EPS) rose significantly to JPY 225.88.

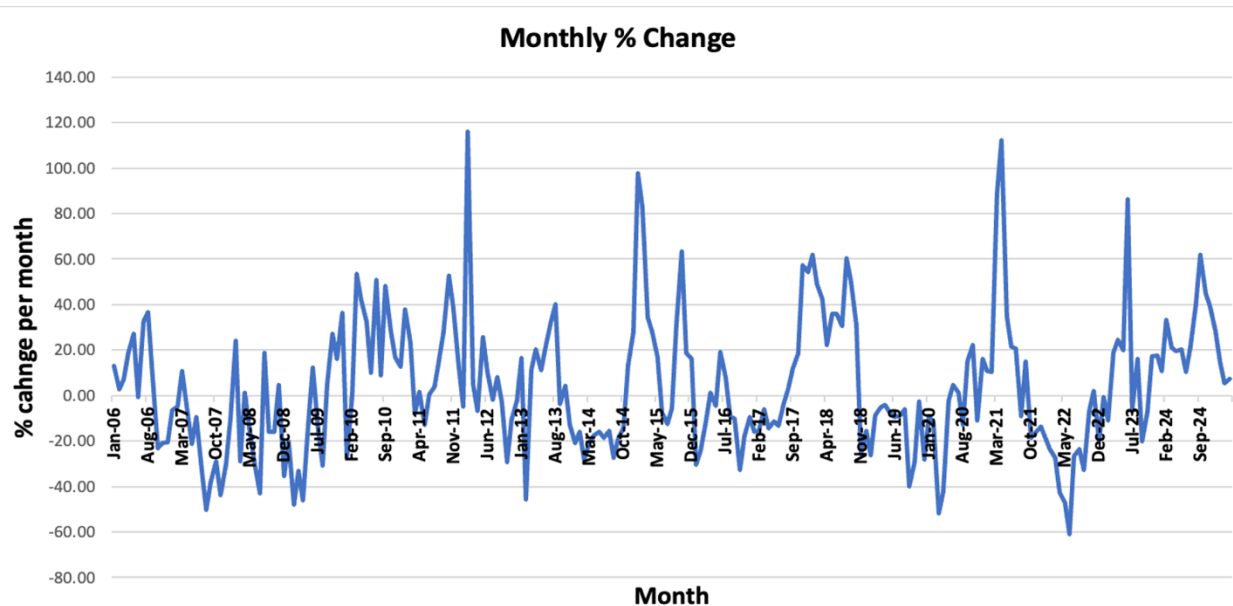
Higher cash reserves and financial service receivables were the main drivers of the JPY 5,104.0 billion increase in total assets as seen from the balance sheet. Increased financing liabilities were a major factor in the JPY 3,600.5 billion increase in liabilities. With the help of currency gains and retained earnings, equity increased by JPY 1,503.5 billion. Honda's strong financial position demonstrates its ability to sustain its investments in hybrid and electric cars as well as its international expansion initiatives.

Monthly Descriptive Analysis of Honda Pilot:

The graph of Monthly Units Sold shows how many units were sold each month across multiple years. When looking at the line chart of the monthly descriptive analysis of Honda Pilot, we can clearly see that the number of units sold goes up and down from one month to the next. Some months have very high numbers, while others have lower numbers.

These changes are not the same every year. There is no regular pattern that repeats, but overall, we can tell that the number of units sold changes frequently. The biggest increases or drops do not happen in the same month every year, which means the monthly sales are not predictable or stable. This makes it harder to know exactly what to expect for sales in any given month.

To estimate the monthly growth rate each year, we compared the number of units sold in the same month but across different years. For example, we compared January 2005 to January 2006. We used the following formula to calculate the percentage growth: $\text{Annual growth} = (\text{units sold in 2006} - \text{units sold in 2005} / \text{units sold in 2005}) \times 100$. We did this for every month, such as comparing February 2005 to February 2006, March 2005 to March 2006, and so on. This helps us see how much the sales increased or decreased for each specific month from one year to the next. A new table was created to show all of these percentage changes, with one row for each month and one column for each year-to-year comparison.



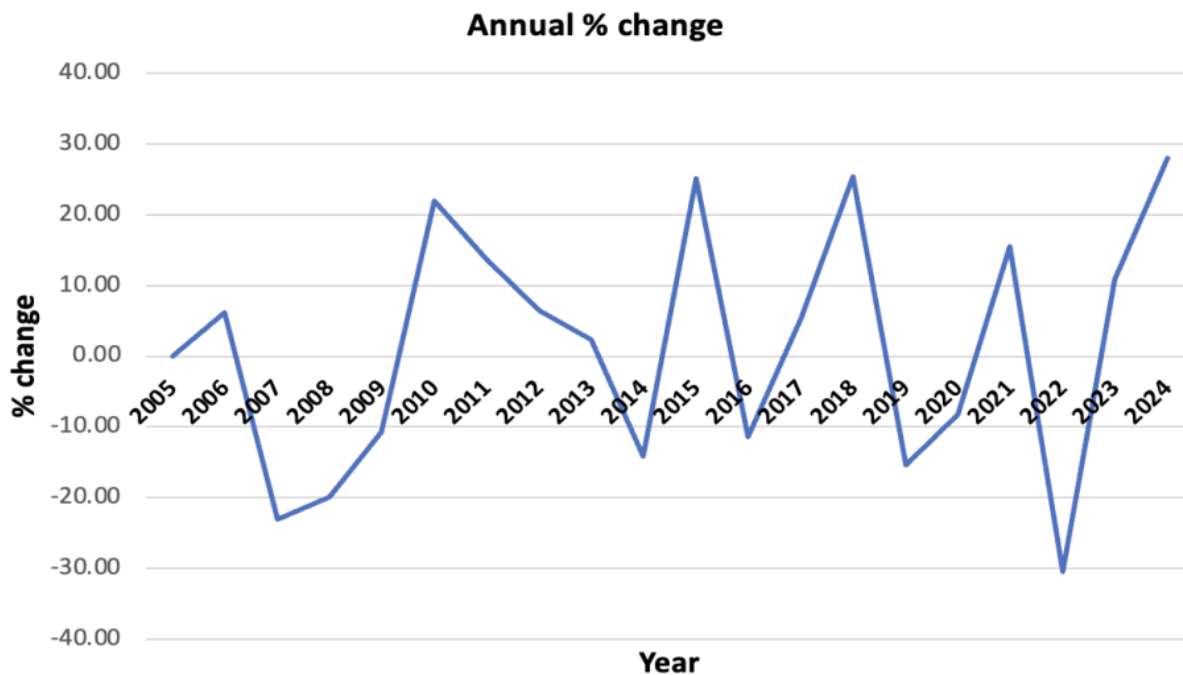
Looking at this line graph of annual monthly growth, we can see that it shows how much each month's sales went up or down compared to the same month in the previous year. The chart 6 shows that some months had a big increase in sales while others had a decrease. These changes are different every year. There is no clear pattern that repeats over time. Some months might have good growth in one year and a drop the next year. This means that Honda Pilot's growth for each month changes depending on what happened during that specific year, such as business decisions, customer interest, or competition, and many other factors.

Annual Descriptive Analysis of Honda Pilot:

The table and linear graph of total units sold annually show how the number of Honda Pilot vehicles sold each year changed over time. From the chart, we can see that most years show increases in total sales, but there are a few years where sales clearly declined. Two important drops happened between 2008 and 2009 and again between 2019 and

2020. Between 2008 and 2009, there was a noticeable drop in total units sold. This happened during the global financial crisis. In 2008, many people lost their jobs, banks failed, and the economy suffered a major recession. Because of this, fewer people were able to afford big purchases like cars, which led to a decline in sales for many car companies, including Honda. A second drop happened between 2019 and 2020. This was the period when the COVID-19 pandemic began. Lockdowns, supply chain problems, factory shutdowns, and lower consumer spending all impacted the car industry. Many dealerships closed for part of the year, and fewer people were shopping for vehicles. These conditions likely caused the significant decline in total units sold during that time.

Overall, while the graph shows long-term growth in some periods, these two specific years (2009 and 2020) stand out as clear declines due to major global events that affected the entire economy. To calculate how much the company's total sales changed each year, we used the same formula we used for the monthly descriptive analysis. It's the formula of the percentage change which is: $(\text{units sold in 2006} - \text{units sold in 2005} / \text{units sold in 2005}) \times 100$. This calculation helped us identify the exact percentage drop in 2009 and 2020. A new table was created to show how much sales increased or decreased each year using this formula.



Now, looking at this line graph of annual growth percentages, we can see that it shows that while there were strong years of growth, two major dips are visible in 2009 and 2020. The 2009 decline reflects the economic crash that affected people's ability to buy cars. The 2020 decline reflects the pandemic, which caused a sudden stop to normal business operations and customer spending. These two years are examples of how external events can strongly affect a company's performance. Other years show smaller ups and downs, but none are as significant as the drops during these two global crises.

Forecasting Analysis:

- **Moving average forecast:** to forecast sales for April 2025 using the Moving Average method, we tested different K values from 2 to 12. Each K value produced a different

forecast, and we compared them using both Mean Absolute Deviation (MAD) and Root Mean Square Error (RMSE) to find the most accurate model.

	K value										
	2	3	4	5	6	7	8	9	10	11	12
RSME	2,440	2,357	2,306	2,320	2,376	2,357	2,337	2,343	2,365	2,340	2,305
MAD	1,778	1,755	1,753	1,774	1,832	1,826	1,797	1,812	1,824	1,829	1,804

After looking at this data, we determined that the minimum RSME was K=12, which provided a forecast of 11,986 units sold, and the minimum MAD was K=4, which gave us a forecast of 11,548 units sold. This means that both K=4 and K=12 are optimal numbers of K.

- **Exponential smoothing forecast:** for the Exponential Smoothing model, we evaluated alpha levels from 0.1 to 0.9 to identify the most accurate smoothing factor. Just like the previous forecasting method, each level of alpha produced a different forecast, and we compared them using both Mean Absolute Deviation (MAD) and Root Mean Square Error (RMSE) to find the forecast sales for April 2025.

	Alpha value								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
RSME	2,336	2,242	2,224	2,238	2,270	2,313	2,369	2,435	2,514
MAD	1,800	1,710	1,679	1,664	1,661	1,668	1,684	1,717	1,769

After revising the forecast values, our team found that alpha=0.3 had the lowest RSME, which showed a forecast of 2,224 units sold, and alpha=0.5 had the lowest MAD, which gave us a forecast of 1,661 units sold. In this case, both alpha=0.3 and alpha=0.5 are optimal levels of alpha.

References

HONDA MOTOR CO., LTD. (2024, May 10). *HONDA MOTOR CO., LTD. REPORTS CONSOLIDATED FINANCIAL RESULTS FOR THE FISCAL YEAR ENDED MARCH 31, 2024*.
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